OUR TEACHING TEAM

Shaojing Fan (Instructor)

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Location:

- E4-05-26 (near ECE general office)

Wang Si (Instructor)

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Leng Yunze (GA---conducts Python Clinic Sessions, handles assignment Q&A and grading)

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OVERVIEW OF COURSE CONTENTS

- **Introduction (Shaojing)**

- What is AI
- Applications of AI
- AI agent

- **Search (Shaojing)**

- Uninformed search algorithms: breadth-first, depth-first, uniform-cost(Dijkstra's algorithm)
- Informed search algorithms: greedy best-first, A*
- Applications

- **Optimisation (Shaojing)**

- Linear programming
- Convex problems
- Applications

- **Machine learning (Wang Si)**

- Supervised and unsupervised learning: regression, classification, clustering
- Neural networks and deep learning
- Applications

- **Knowledge representation (Wang Si)**

- Knowledge Representation and Reasoning
- Propositional Logic
- Applications

- **Ethical considerations (Shaojing)**

- Bias in AI
- Privacy concerns
- Societal impact

COURSE PLAN

mechanistic, rule-based forms of intelligence

Search Optimisation

lower-Level/foundational intelligence

adaptive, data-driven forms of intelligence

Machine learning

mid-level/adaptive intelligence

abstract, reasoning-based forms of intelligence

Knowledge representation

higher-level/reasoning Intelligence

Grading:

In-Class Quizzes	20%
Projects	20%
Assignments	20%
Final exam	40%

Class Venue:

- **Monday**: LT4
- **Friday**: LT7

Four In-Class Quizzes

No.	Test Content	Grade %	Week	Date
Quiz 1	Search	5%	Week 3	29 Aug 2025
Quiz 2	Optimization	5%	Week 5	12 Sep 2025
Quiz 3	Machine learning	5%	Week 9	17 Oct 2025
Quiz 4	Knowledge representation	5%	Week 11	31 Oct 2025

- Mode: Digital exam conducted via Examplify
- Open-book exam: You may bring any printed or digital materials, but internet access is not allowed. You will have 2 hours to complete the exam.

Gentle reminder: Please lock the dates in your calendar. ***Only students with a valid MC will be excused****, and a make-up quiz will be arranged.*

Reference of Examplify:
https://nus.atlassian.net/wiki/spaces/DAstudent/pages/22511650/Getting+Started

One Project (Individual):

Machine learning 20%

- This is an open-book project. You will be given four weeks to complete it, and all coding must be done in Python.

- Late submission policy:
    - 0–12 hours late: 80% of your grade
    - 1 day late: 50% of your grade
    - 2 days late: 30% of your grade
    - After 2 days: 0

Four After-Class Assignments

Search (2 assignments)	4% + 6%
Optimization (1 assignments)	5%
Knowledge Representation (1 assignments)	5%

- Open-book; you will have 2–3 weeks to complete each assignment.
- Code will be autograded, with feedback provided on a subset of test cases.
- Late submission policy:
  - 0–12 hours late: 80% of your grade
  - 1 day late: 50% of your grade
  - 2 days late: 30% of your grade
  - After 2 days: 0

Python Clinic Sessions

Online Python Clinic Session 1 for EE2213:

- Slot 1: Thursday, 21 Aug, 9:00–10:00 PM
- Slot 2: Monday, 25 Aug, 3:00–4:00 PM

Online Python Clinic Session 2 for EE2213:

- Slot 1: Monday, 29 Sep, 3:00–4:00 PM

***Python clinic sessions are optional but highly recommended. They cover fundamental Python concepts and skills to support you in completing assignments and coding-related exam/quiz questions.**

Final Exam (40%)

Goal: To assess your ability to apply knowledge to new problems, rather than just recall facts.

Mode: Digital exam conducted via Examplify

Open-book exam: You may bring any printed or digital materials, but internet access is not allowed.

What is Artificial Intelligence (AI)?

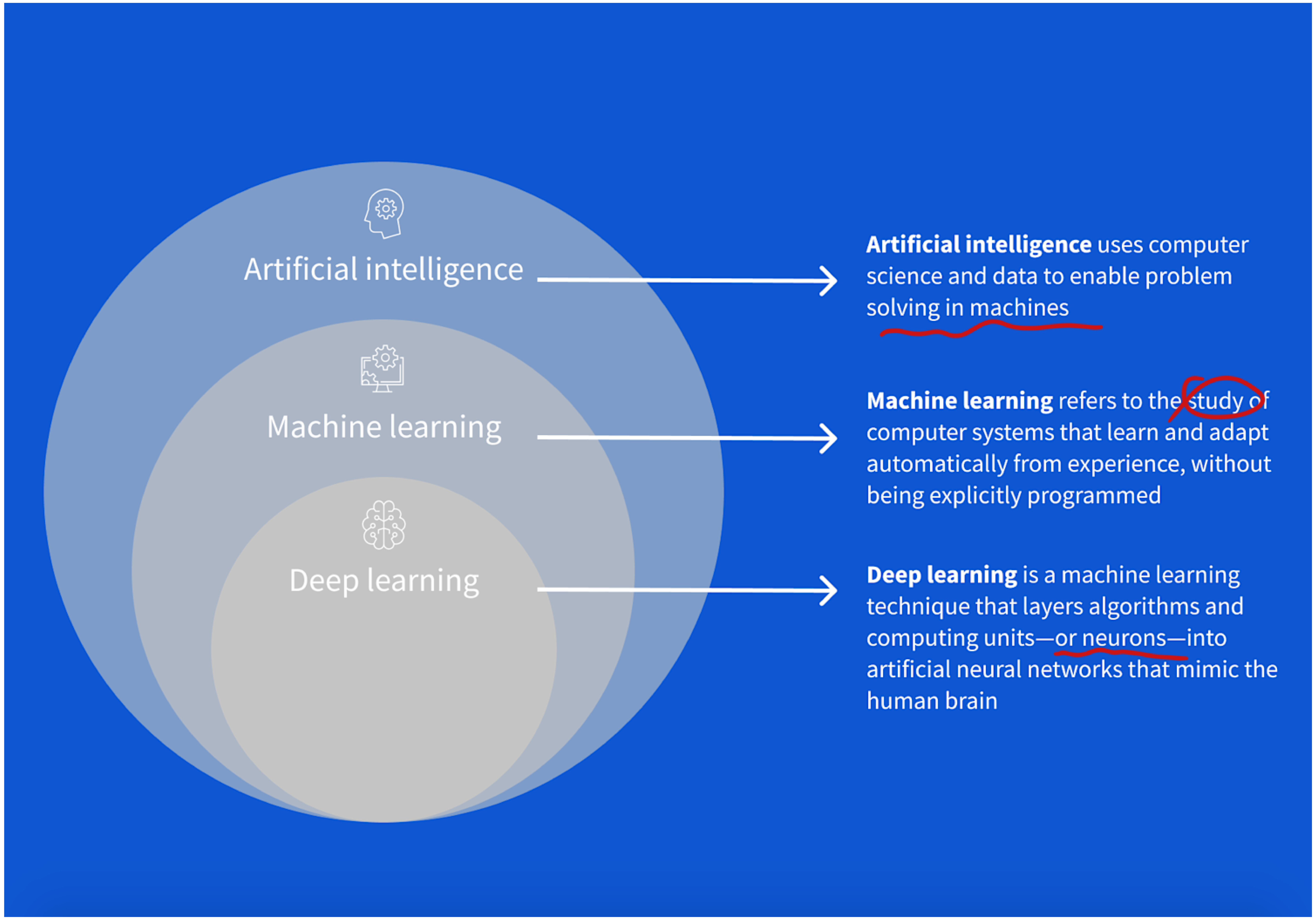
Artificial intelligence is the study of agents that perceive their environment and take actions that maximize their chances of achieving their goals.

— Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.)

The science and engineering of making intelligent machines, especially intelligent computer programs.

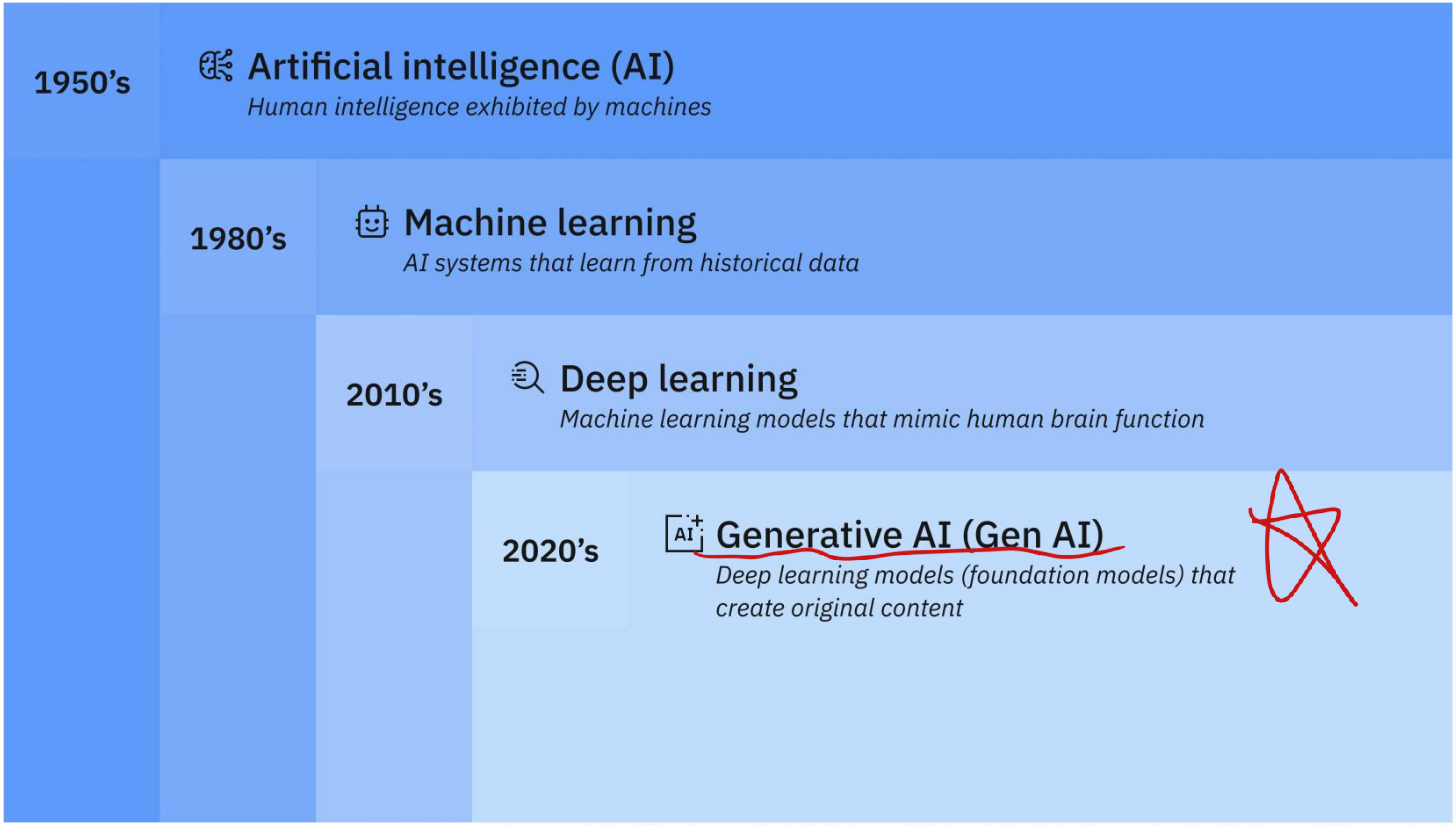
— John McCarthy (1956), the "Father" of AI, who coined the term Artificial Intelligence

AI, Machine Learning, and Deep Learning



Reference: https://www.coursera.org/articles/ai-vs-deep-learning-vs-machine-learning-beginners-guide

AI, Machine Learning, Deep Learning, and Generative AI

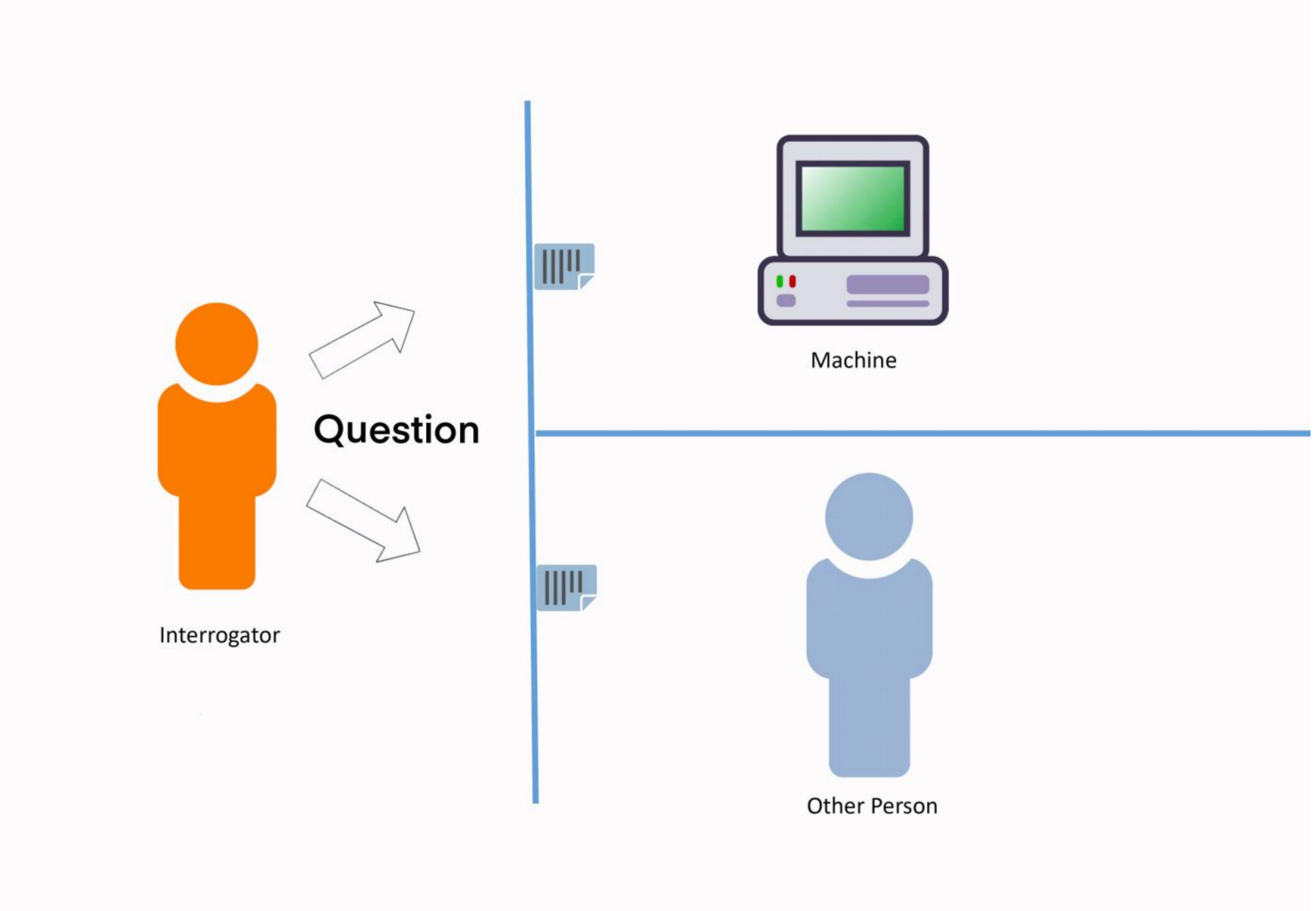


Credit: https://www.ibm.com/think/topics/artificial-intelligence

Alan Turing and Turing Test

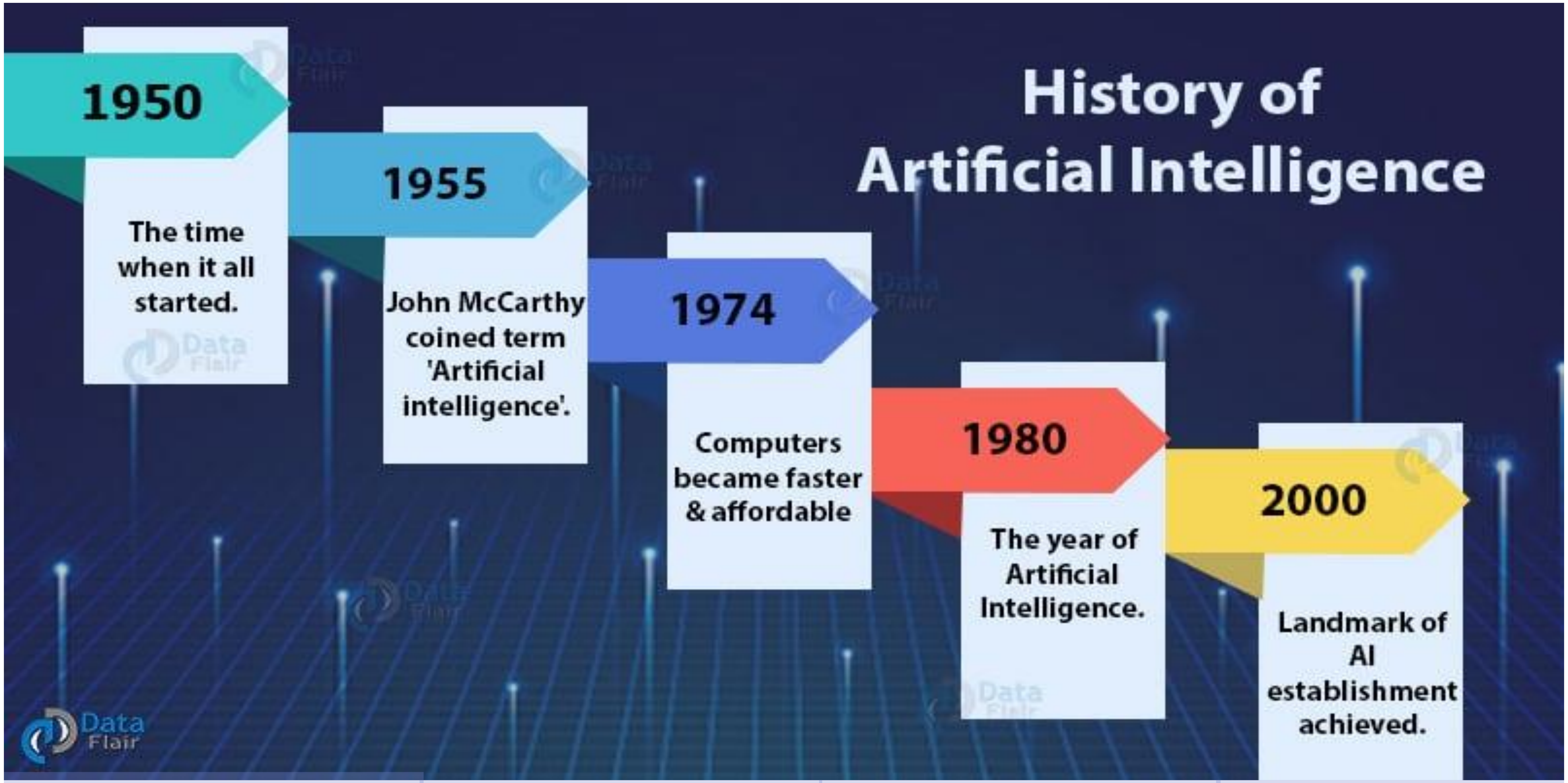
"A computer would deserve to be called intelligent if it could deceive a human into believing that it was a human."

— Alan Turing



Reference: Jones, Cameron R., and Benjamin K. Bergen. "Large language models pass the turing test." *arXiv preprint arXiv:2503.23674* (2025).

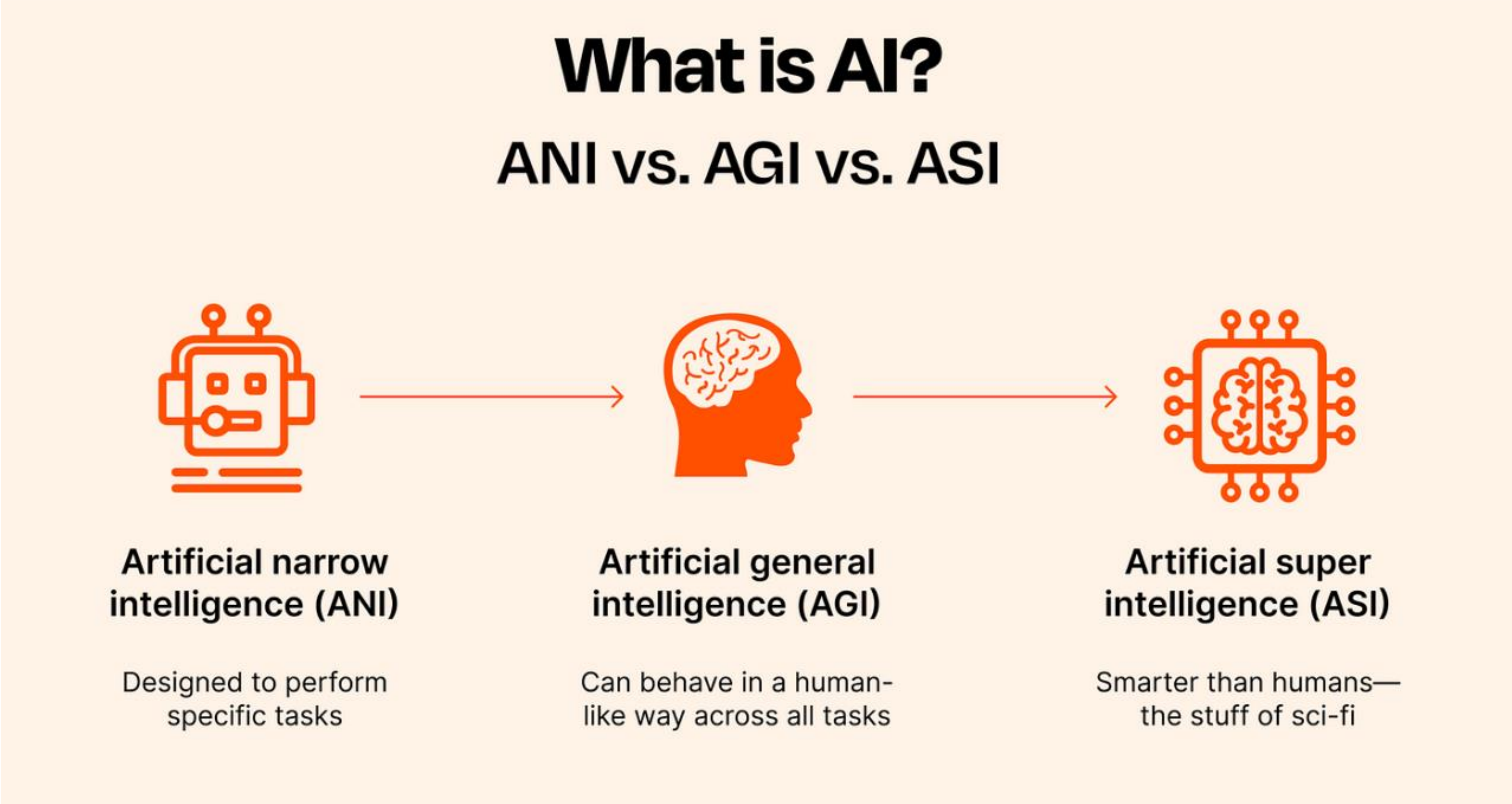
A Brief AI History



Birth of AI (1950s)	**Early Development (1960s-70s)**	**Resurgence (1980s-2000s)**	**Modern AI (2010s-Present)**
Turing Test (1950)	ELIZA chatbot	Rule-based systems	Deep learning
Dartmouth Conference (1956)	First AI Winter	Deep Blue (1997)	Generative AI

Reference: https://data-flair.training/blogs/history-of-artificial-intelligence/

AI in Future



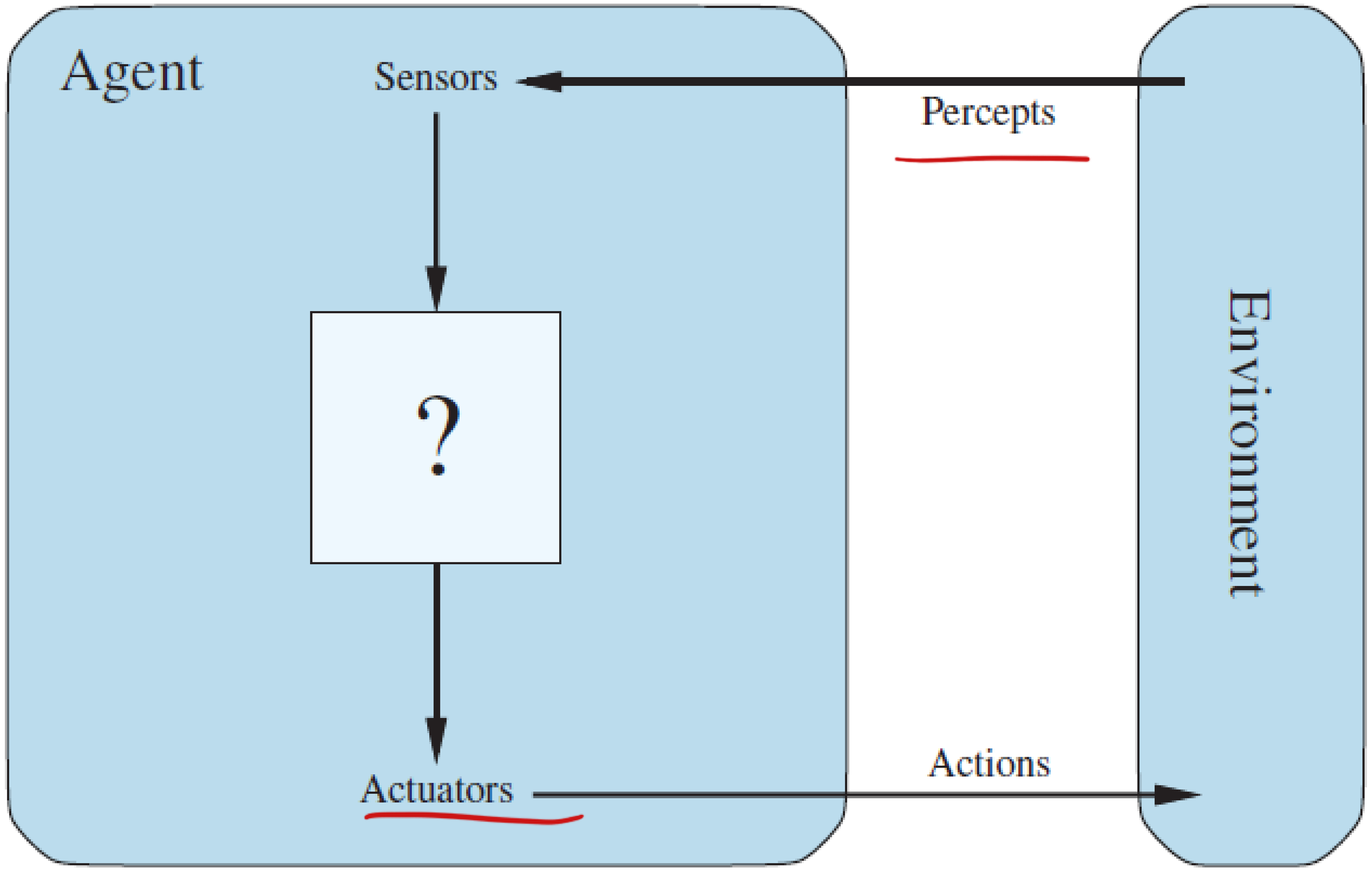
Nick, Bostrom. "Superintelligence: Paths, dangers, strategies." 2014,

Definition of Agent

An **agent** is anything that **perceives** its environment through **sensors** and can **act** on its environment through **actuators**

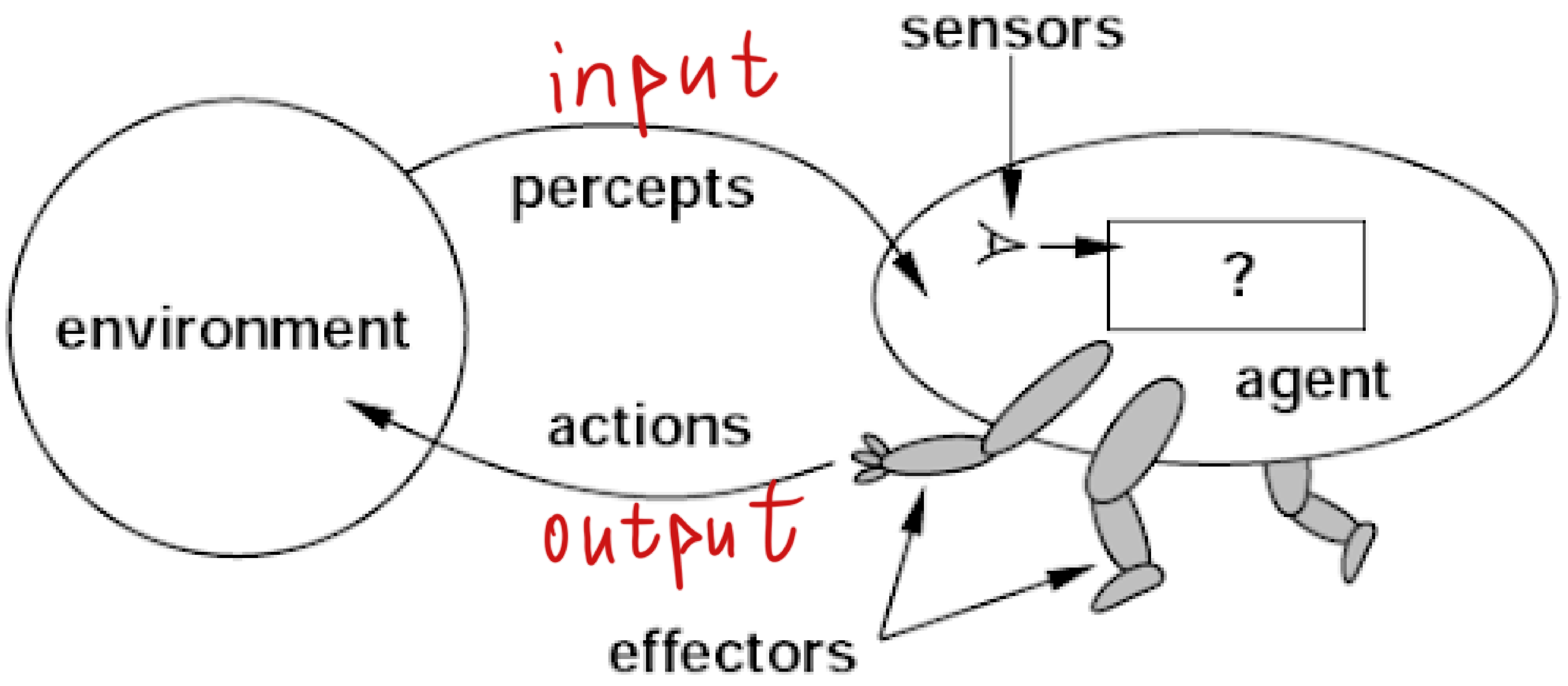
A **percept** is the agent's perceptual inputs at any given instance

An **actuator** is the part of an agent that allows it to take action in the environment based on its decisions.



Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P54 – P57.

Agents and Environments

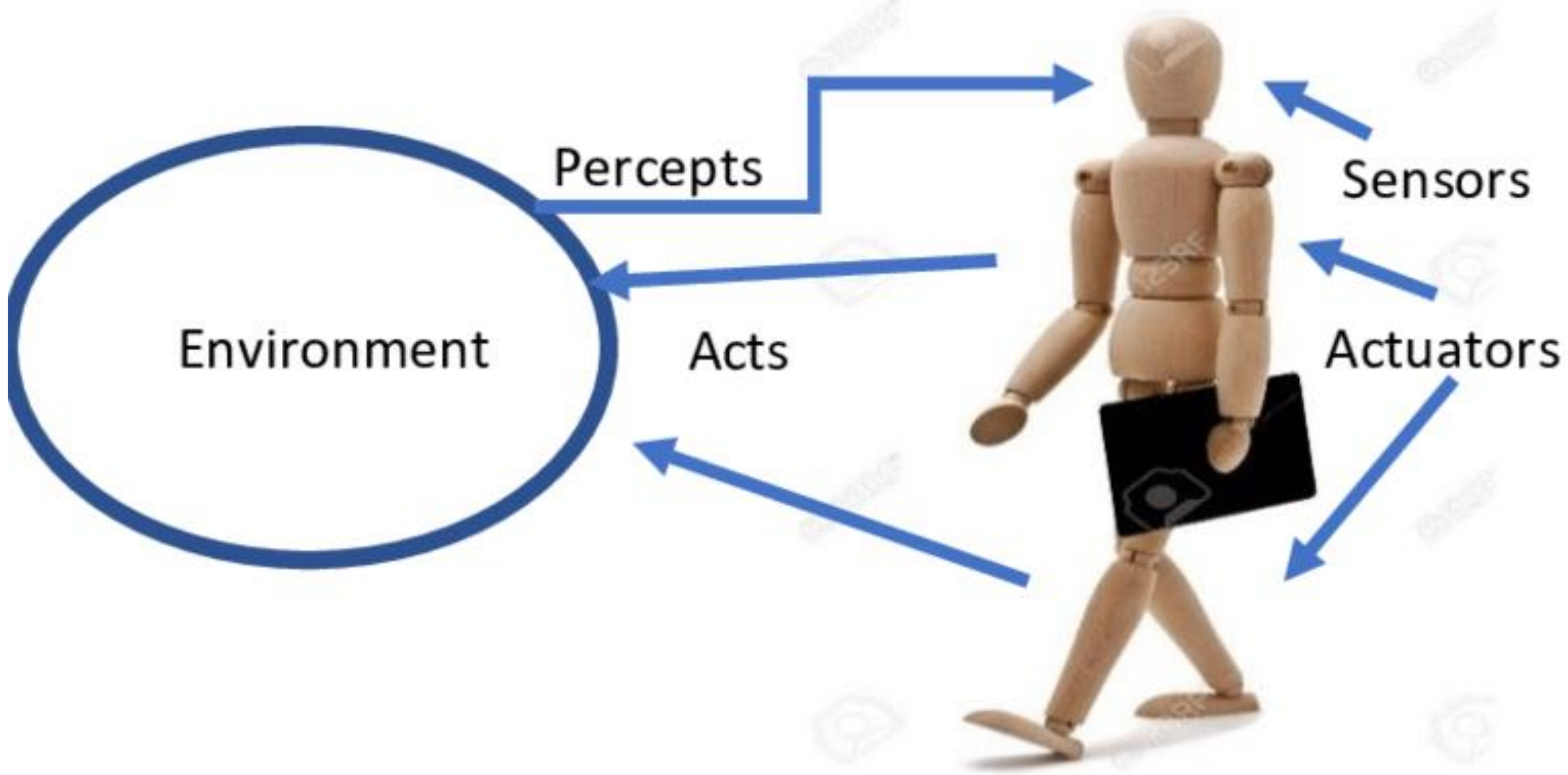


An agent is specified by an *agent function* $f: P \rightarrow a$ that maps a sequence of percept vectors P to an action a from a set A:

$$P = [p_0, p_1, \dots, p_t]$$
$$A = [a_0, a_1, \dots, a_t]$$

Examples of Agents

An ***agent*** is anything that can be viewed as
- ***perceiving*** its ***environment*** through ***sensors*** and
- ***acting*** upon that environment through ***actuators***



Human agent:
- Sensors: eyes, ears, …
- Actuators: hands, legs, mouth, …

Robotic agent:
- Sensors: cameras and infrared range finders
- Actuators: various motors

Agents include humans, robots, softbots, thermostats…

Performance Measure

How do we know if an agent is acting rationally?

- Informally, we expect that it will do the right thing in all circumstances

How do we know if it's doing the right thing?

We define a ***performance measure***:

- An objective criterion for success of an agent's behavior
- Given the evidence provided by the percept sequence

E.g., performance measure of a vacuum-cleaner agent could be amount of dirt cleaned up, amount of time taken, amount of electricity consumed, amount of noise generated, etc.

Rational Agents

Rational Agent (initial definition):

- For each possible percept sequence *P*
- a rational agent should select an action *a*
- to maximize its performance measure

Rational Agent (revised definition):

- For each possible percept sequence *P*
- a rational agent should select an action *a*
- to maximize the **expected value** of its performance measure

It doesn't have to know what the actual outcome will be.

Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P58 – P60.

Task Environment

To design a rational agent, we need to specify a **task environment** --- the setting/context in which the agent will operate and make decisions

- the problem specification the agent is meant to solve

PEAS: to specify a task environment

- ***P***erformance measure
- ***E***nvironment
- ***A***ctuators
- ***S***ensors

Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P60 – P61.

PEAS: Example

Agent: automated vehicle

*P*erformance measure
- safe, fast, legal, comfortable, maximize profits

*E*nvironment
- roads, other traffic, pedestrians, customers

*A*ctuators
- steering, accelerator, brake, signal, horn

*S*ensors
- cameras, LiDAR, speedometer, GPS

Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P61.

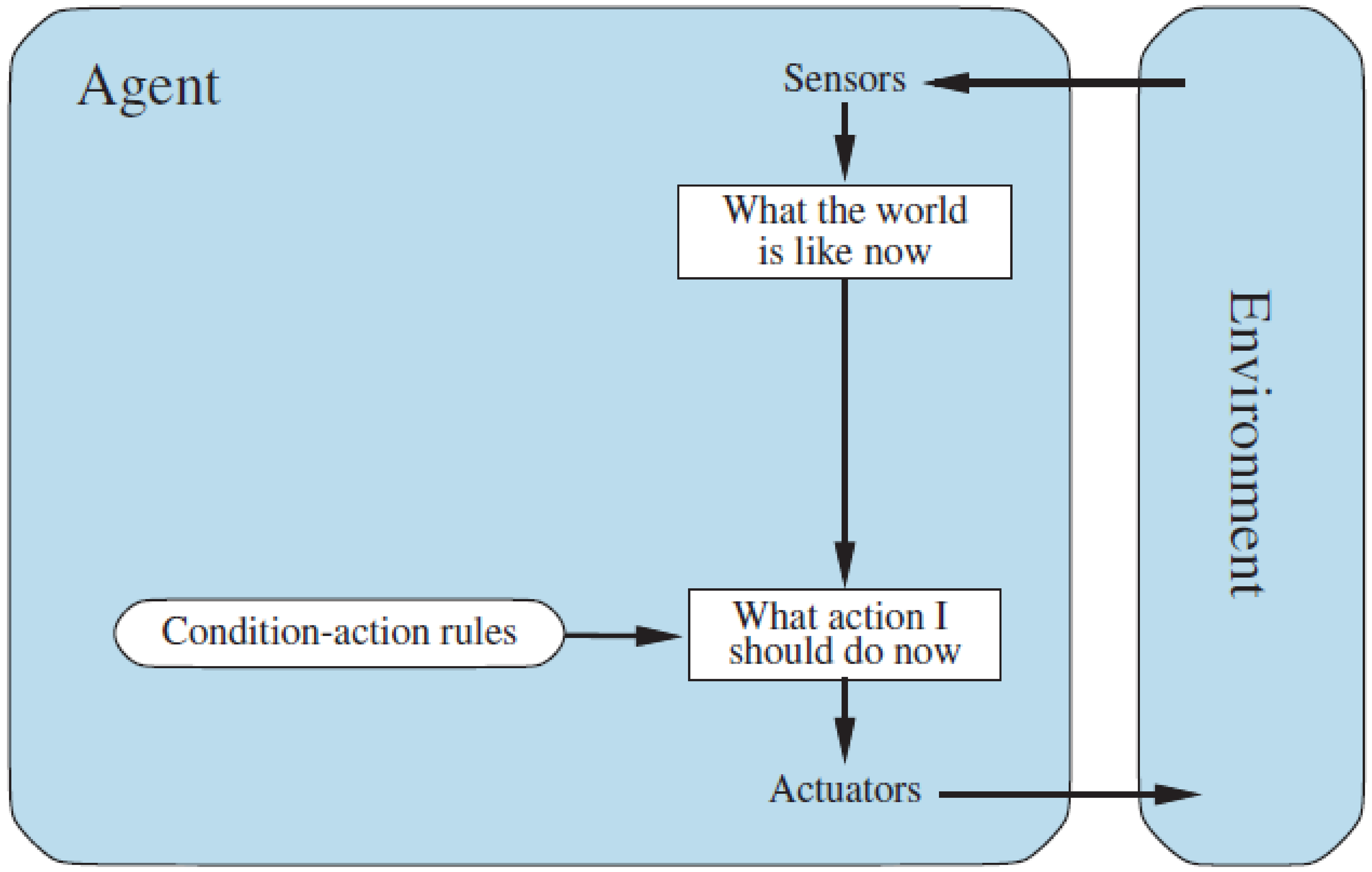
Simple Reflex Agents

- **A Reflex Agent**
  - Selects action based on the current percept.
  - Directly map states to actions.

- Operate using *condition-action* rules.
  - **If** (condition) **then** (action)

- Example:
  - A vacuum cleaner robot that reacts to dirt.
  - **If** *(current-location-is-dirty) then (suck-dirt)*

- Problem: no notion of goals
  - It doesn't plan where to clean next or remember cleaned areas.
  - may repeatedly visit clean spots and miss others.



Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P67 – P69.

From Reflex to Problem Solving Agents

- **Reflex Agents**
  - Directly map states to actions.
  - No notion of goals. Do not consider action consequences.

$x \longrightarrow \boxed{f} \longrightarrow$ single action y

- **Problem Solving Agents**
  - Adopt a goal
  - Consider future actions and the desirability of their outcomes
  - Solution: a sequence of actions that the agent can execute leading to the goal.
  - Today's focus.

$x \longrightarrow \boxed{f} \longrightarrow$ **action sequence** $(a_1, a_2, a_3, a_4, \ldots)$

Traditional AI Systems vs. Agentic AI

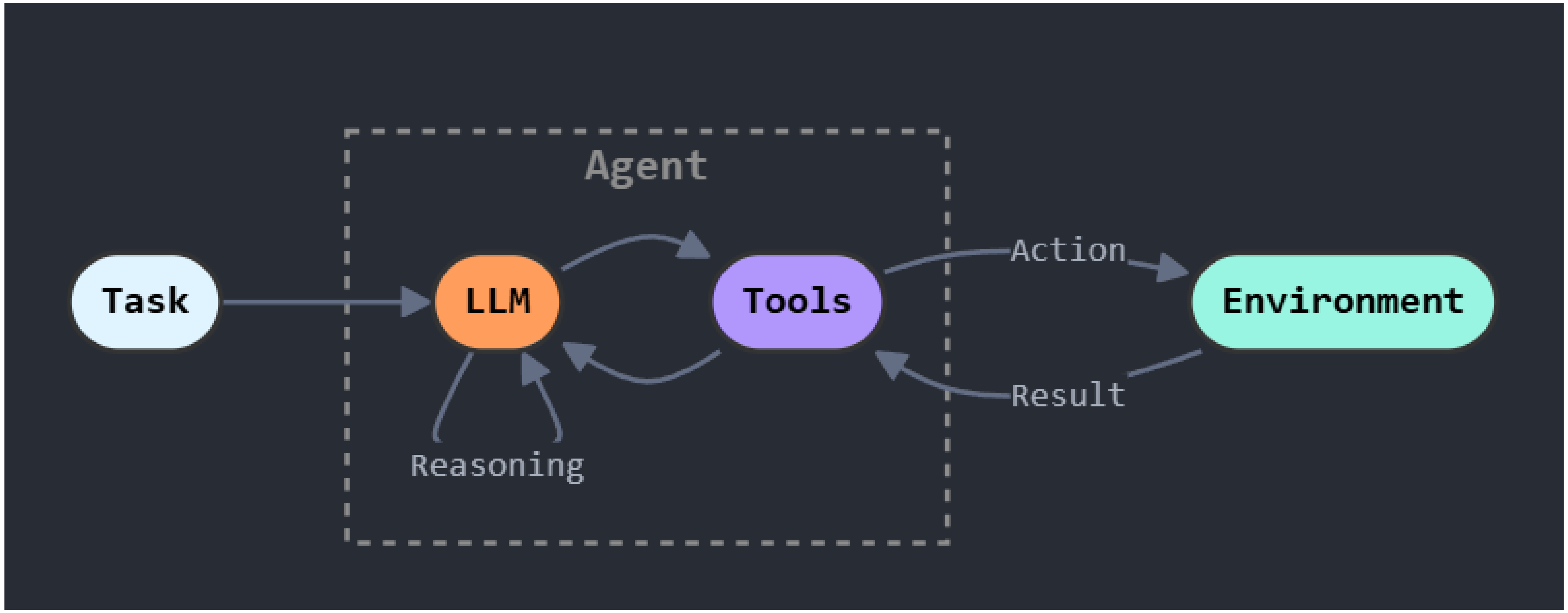
Nowadays

Features	Traditional AI	Agentic AI
Decision-Making	Follows rules, no autonomy	Independent goal-setting & action
Adaptability	Limited, needs reprogramming	Learns & adjusts continuously
Reactivity	Waits for input	Predicts and acts proactively
Execution	Requires step-by-step guidance	Self-directed execution

Credit: https://www.civo.com/blog/what-is-agentic-ai

ReAct AI Agent

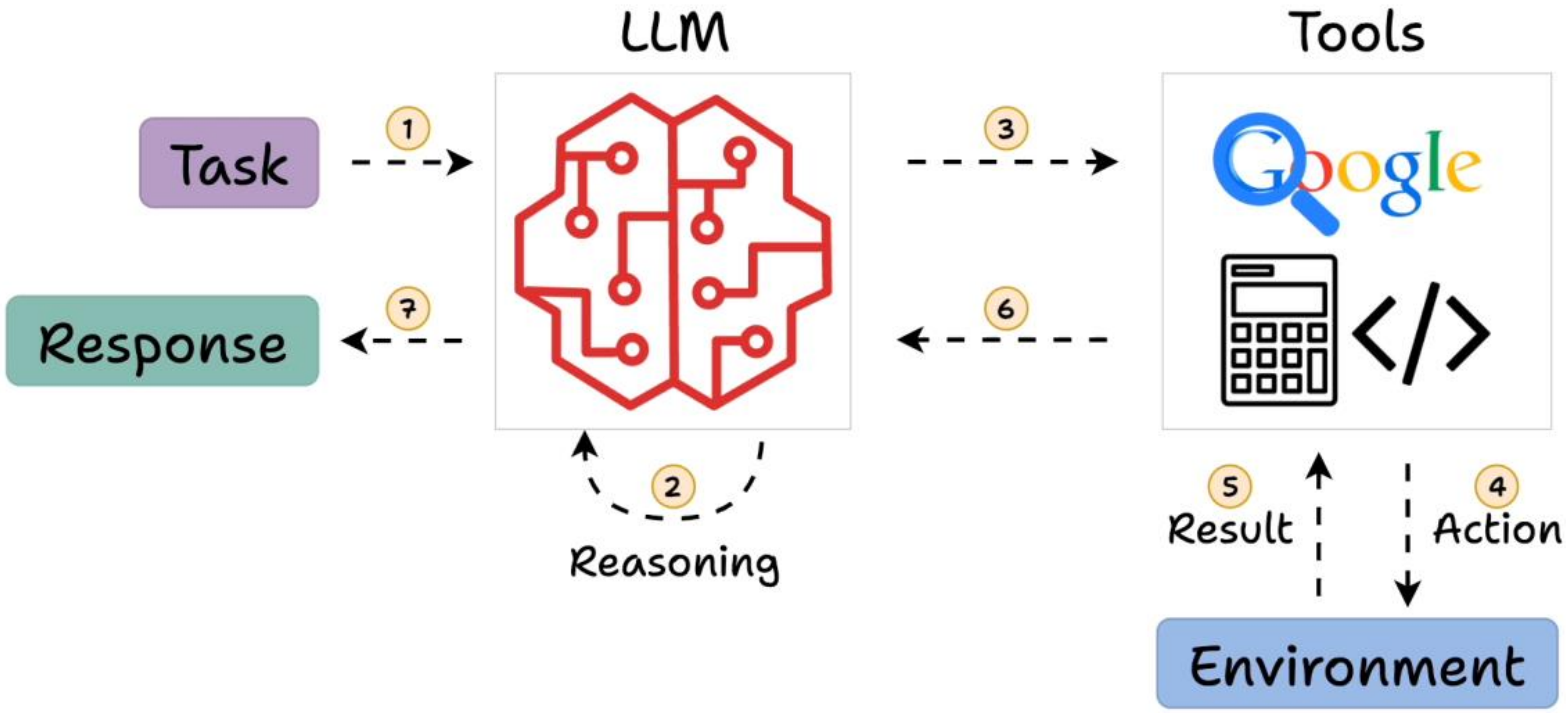
A **ReAct agent** (short for *Reasoning and Acting*) is a framework that enables LLMs to alternately generate explicit reasoning steps (thoughts) and perform actions (like API or tool calls) in a loop—synergizing internal reasoning with external execution to solve complex, multi-step tasks



Reference:
https://medium.com/%40gauritr01/part-1-react-ai-agents-a-guide-to-smarter-ai-through-reasoning-and-action-d5841db39530
https://www.youtube.com/watch?v=F8NKVhkZZWI&list=PL1SWNKG9Fr7kEWZ91kFWw2R3CmXqH7jx1&index=15
https://medium.com/@DanGiannone/demystifying-ai-agents-react-style-agents-vs-agentic-workflows-cedca7e26471

ReAct AI Agent

I am in London, do I need an umbrella today?



- Understand the query (language understanding)
- Select the tools it would need to generate a response (the weather API).
- Gather the weather report.
- Understand if it’s going to rain and produce an answer accordingly.

Reference: https://blog.dailydoseofds.com/p/intro-to-react-reasoning-and-action
https://www.youtube.com/watch?v=F8NKVhkZZWI&list=PL1SWNKG9Fr7kEWZ91kFWw2R3CmXqH7jx1&index=15